

SAI HARSHITH VARMA PERICHERLA

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Summary

Data Engineer with 2+ years' experience in **ETL/ELT pipelines, cloud data warehouses, and lakehouse architecture**, processing **2+ TB/day** with **99.8% reliability** and improving data efficiency by **45%**. Skilled in **PySpark, Apache Spark, Airflow, Kafka, AWS, Azure, GCP, SQL, Python, dbt, Tableau, and Power BI**, delivering **50+ dashboards** and reducing manual reporting by **50%**. Experienced in **CDC, dimensional modeling, data governance, RBAC/IAM, and GDPR/SOC 2 compliance**, enabling **scalable analytics and actionable insights** across enterprise environments.

Technical Skills

Programming & Scripting: SQL, Python (Pandas, NumPy, scikit-learn), Java, Scala, Bash, Shell Scripting

Data Engineering & Architecture: ETL/ELT, Data Warehousing, Data Lakes & Lakehouse Architecture, Change Data Capture (CDC), Data Modeling (Dimensional & Relational), Performance Tuning, Batch & Streaming Pipelines

Big Data & Streaming: Apache Spark (PySpark, Spark SQL, Structured Streaming), Apache Kafka, Kafka Streams, Hadoop, Apache Flink, Hive

Workflow & Orchestration: Apache Airflow, Azure Data Factory, AWS Glue, Prefect, Cron, Dagster, Apache Beam

Cloud Platforms: AWS (S3, Redshift, EMR, Athena, Glue, Lambda, IAM), Microsoft Azure (ADLS, Synapse Analytics, Databricks, Data Factory), Google Cloud Platform (BigQuery, Dataflow, Dataproc, Cloud Composer)

Databases & Storage Systems: PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Cassandra, Snowflake, DynamoDB, Redis, HBase

Modern Data Stack & Analytics: dbt (Core & Cloud), Tableau, Power BI, Looker Studio, Semantic Modeling, Metrics Layer, BI Reporting, Data Visualization

Data Governance, Quality & Security: Data Governance, Data Quality Frameworks, Data Lineage, Metadata Management, Master Data Management (MDM), Role-Based Access Control (RBAC), IAM, Encryption, HIPAA, GDPR, SOC 2

Lakehouse & Table Formats: Delta Lake, Apache Iceberg, Apache Hudi, ACID Transactions, Time Travel, Versioned Data

DevOps, MLOps & Monitoring: Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Jenkins, GitLab CI), Git, Prometheus, Grafana, ELK Stack, MLflow

Integration & Data Formats: REST APIs, SOAP APIs, JSON, XML, Parquet, Avro, ORC, Protobuf, Flat Files

Tools & Methodologies: Linux/Unix, Jupyter Notebook, VS Code, Advanced Excel, Agile, Scrum, DataOps, MLOps, SDLC, JIRA, Confluence

Professional Experience

State Street

Role: Data Engineer

NJ, USA

Feb 2026 – Present

- Designed and optimized **30+ scalable ETL/ELT pipelines** using PySpark, Airflow, and AWS Glue, improving data processing efficiency by **45%** and reducing latency by **35%**.
- Built and maintained cloud-based data warehouses on AWS Redshift and S3, supporting **50+ analytical dashboards** and enabling **real-time insights for 10+ business units**.
- Implemented CDC and dimensional data models, increasing data accuracy by **40%** and reducing reporting discrepancies by **28%**.
- Automated batch and streaming workflows using Apache Kafka and Spark Structured Streaming, processing **2+ TB of data daily** with **99.8% pipeline reliability**.
- Strengthened data governance and security by enforcing RBAC, IAM policies, and encryption standards, ensuring **100% compliance** with GDPR and SOC 2 regulations.

Mphasis

Role: Data Engineer

Hyderabad, India

May 2021 – Dec 2023

- Developed and deployed **15+ data pipelines** using Python, SQL, and Apache Spark, reducing manual reporting efforts by **50%**.
- Supported migration of on-premises databases to AWS and Azure platforms, improving system availability by **30%** and lowering infrastructure costs by **20%**.
- Optimized SQL queries and indexing strategies, enhancing query performance by **35%** across enterprise reporting systems.
- Integrated REST APIs and structured data sources into centralized data lakes, enabling **25% faster analytics delivery**.
- Collaborated with cross-functional teams using Agile methodologies, accelerating project delivery timelines by **22%**.

Education

Master of Science in Business Analytics | University of New Haven

Dec 2025

Projects

- Fraud Detection in Financial Options Trade Data:** Increased fraud detection accuracy by 40% on options trade data by implementing Variational encoders in Keras and TensorFlow. Achieved a Matthews Correlation Coefficient (MCC) of 0.86 by minimizing false predictions through feature engineering and hyperparameter tuning, enhancing model reliability for real-time fraud prevention.
- Customer Analytics for a Retail Store Giant:** Analyzed customer behavior and segmented purchasing patterns to reveal high-performing product groups, leading to a 38% improvement in promotion ROI. Identified the pain points in sales and inventory by KPI tracking. Created Power BI dashboards and scalable SQL queries that standardized definitions and reduced manual reporting time by improving decision cadence.